

# SinusitisTherapy

DTRules rules reference — data model & decision tables

# How this rule set is built

A DTRules application is two artifacts authored in spreadsheets: an **Entity Data Dictionary** that defines the data, and a set of **decision tables** that define the logic. This document lists both, plus how they drive an interactive interview.

## The Entity Data Dictionary (EDD) — what it contains

The EDD is the data model. It declares the **entities** (things the rules reason about, e.g. a patient) and, for each, its **attributes**: a name, a type (string, integer, decimal, boolean, date, an entity reference, or an array), an optional default, and whether the field is an *input* the rules read or an *output* they produce. The decision tables may only reference fields declared here — the EDD is the shared vocabulary.

An attribute can also be marked **collected**: rather than arriving as bulk input, the engine asks the user for it. Collected fields carry question metadata — a prompt, a type (`multiple_choice`, `ascii`, `number`, `date`), choice options, and for numbers a lab-style normal range.

## Decision tables — what they contain

Each decision table is a grid. The top rows are **conditions** (when something is true) and the bottom rows are **actions** (what to do); every numbered **column is one rule**. A condition cell is **Y** (must be true), **N** (must be false), or **–** (don't care). An action cell is marked when that action fires for the column. Each condition and action is written in DTRules' Expression Language (EL), shown in the grid.

- **Policy** — **FIRST** fires the first matching column, **ALL** fires every matching column in order, **BALANCED** expects exactly one column to match.
- **Initial actions** run once before the columns; **contexts** push an entity (e.g. "for all active medications") so the table evaluates against each.
- Tables call other tables with `perform`, composing larger logic from small, readable pieces.

## The interview UI is automatic

There is no hand-built form. When the engine *executes the decision tables* and a rule reads a collected field that hasn't been supplied, it pauses and asks that field's question — on the command line or in the browser — then resumes. The screens a user sees are a by-product of which fields the tables actually reach: change the tables or the EDD questions and the interview changes with them, with no UI code to update.

# Entity Data Dictionary

The data model the decision tables read and write. ● marks a field the interview collects from the user.

## constants

| Field                 | Type    | Default | I/O   | Asked as | Description                                    |
|-----------------------|---------|---------|-------|----------|------------------------------------------------|
| adult_age             | integer | 18      | input | —        | Age at which a patient is treated as an adult  |
| dose_high_age         | integer | 60      | input | —        | Upper bound of the standard-dose age band      |
| dose_low_age          | integer | 15      | input | —        | Lower bound of the standard-dose age band      |
| dosing_interval_hours | integer | 24      | input | —        | Standard dosing interval in hours              |
| reduced_dose          | integer | 200     | input | —        | Reduced dose in mg                             |
| renal_ccr_threshold   | double  | 50.0    | input | —        | CCr below which renal adjustment applies       |
| renal_pcr_threshold   | double  | 1.4     | input | —        | PCr above which renal adjustment is considered |
| standard_dose         | integer | 500     | input | —        | Standard dose in mg                            |
| therapy_duration_days | integer | 14      | input | —        | Standard therapy duration in days              |

## medication

| Field | Type   | Default | I/O    | Asked as | Description                  |
|-------|--------|---------|--------|----------|------------------------------|
| name  | string | —       | in-out | —        | Drug name (brand or generic) |

## patient

| Field                 | Type                | Default         | I/O    | Asked as                    | Description                                      |
|-----------------------|---------------------|-----------------|--------|-----------------------------|--------------------------------------------------|
| active_medications    | array of medication | —               | in-out | —                           | Medications the patient is currently taking      |
| ● age                 | integer             | 0               | in-out | number in years             | Patient age in years                             |
| ● diagnosis           | string              | Acute Sinusitis | in-out | ascii                       | Working diagnosis, e.g. Acute Sinusitis          |
| ● lean_body_weight    | double              | 0.0             | in-out | number in kg                | Lean body weight in kg (used by the CCr formula) |
| ● pcr                 | double              | 0.0             | in-out | number normal 0.7–1.3 mg/dL | Plasma creatinine level (mg/dL)                  |
| ● penicillin_allergic | boolean             | false           | in-out | multiple_choice (Yes, No)   | True if the patient is penicillin-allergic       |

result

| Field            | Type            | Default | I/O    | Asked as | Description                                                         |
|------------------|-----------------|---------|--------|----------|---------------------------------------------------------------------|
| ccr              | double          | 0.0     | in-out | —        | Service 2 output: creatinine clearance (mL/min)                     |
| dose_mg          | integer         | 0       | output | —        | Service 1 output: single dose in mg                                 |
| duration_days    | integer         | 0       | output | —        | Service 1 output: therapy duration in days                          |
| frequency_hours  | integer         | 0       | output | —        | Service 1 output: hours between doses                               |
| recommended_drug | string          | —       | in-out | —        | Service 1 output: drug of choice                                    |
| renal_adjustment | boolean         | false   | output | —        | True if the dose was reduced for renal impairment                   |
| warnings         | array of string | —       | in-out | —        | Service 3 output: drug-interaction warnings                         |
| rationale        | array of string | —       | in-out | —        | Human-readable rationale for each service decision (explainability) |

# Decision Tables

6 tables. Conditions (top) decide which rule columns apply; actions (bottom) run for the marked columns.

**Y** condition must be true    **N** must be false    – don't care    ✓ action fires

## 1000 · Determine\_Therapy

FIRST

| #      | Condition                                                                                         | 1 |
|--------|---------------------------------------------------------------------------------------------------|---|
| 1      | patient.diagnosis is equal to ignore case "Acute Sinusitis"<br>Always run the full therapy workup | Y |
| ACTION |                                                                                                   | 1 |
| 1      | perform Determine_Creatinine_Clearance<br>Service 2 first: CCr feeds the dosing rules             | ✓ |
| 2      | perform Determine_Medication_And_Dosing<br>Service 1: drug of choice and dosing                   | ✓ |
| 3      | perform Check_Drug_Interactions<br>Service 3: drug-interaction warnings                           | ✓ |

## 2000 · Determine\_Creatinine\_Clearance

FIRST

| #      | Condition                                                                                                                                                               | 1 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 1      | true<br>Always compute CCr                                                                                                                                              | Y |
| ACTION |                                                                                                                                                                         | 1 |
| 1      | set result.ccr = ((140 - patient.age) * patient.lean_body_weight) /<br>(patient.pcr * 72)<br>Cockcroft-Gault style: CCr = ((140 - age) * lean body weight) / (PCr * 72) | ✓ |
| 2      | add "CCr estimated via Cockcroft-Gault from age, lean body weight, and plasma creatinine." to result.rationale<br>Rationale: CCr method                                 | ✓ |

## 3000 · Determine\_Medication\_And\_Dosing

FIRST

| #      | Condition                                                      | 1 |
|--------|----------------------------------------------------------------|---|
| 1      | true<br>Always run                                             | Y |
| ACTION |                                                                | 1 |
| 1      | perform Select_Medication<br>Pick the drug of choice           | ✓ |
| 2      | perform Determine_Dose<br>Pick the dose / frequency / duration | ✓ |

## 3010 · Select\_Medication

### FIRST

| #      | Condition                                                                                                                                                                | 1 | 2 | 3 |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 1      | patient.penicillin_allergic is true<br>Penicillin allergy overrides age (beta-lactam cross-reactivity)                                                                   | Y | – | – |
| 2      | patient.age >= constants.adult_age<br>Adult (>= 18)                                                                                                                      | – | Y | N |
| ACTION |                                                                                                                                                                          | 1 | 2 | 3 |
| 1      | set result.recommended_drug = "Levofloxacin"<br>Fluoroquinolone for penicillin-allergic patients                                                                         | ✓ |   |   |
| 2      | set result.recommended_drug = "Amoxicillin"<br>Adult first-line                                                                                                          |   | ✓ |   |
| 3      | set result.recommended_drug = "Cefuroxime"<br>Pediatric first-line                                                                                                       |   |   | ✓ |
| 4      | add "Levofloxacin selected: fluoroquinolone avoids beta-lactam cross-reactivity in a penicillin-allergic patient." to result.rationale<br>Rationale: penicillin-allergic | ✓ |   |   |
| 5      | add "Amoxicillin selected: first-line beta-lactam for a non-allergic adult." to result.rationale<br>Rationale: adult first-line                                          |   | ✓ |   |
| 6      | add "Cefuroxime selected: first-line for a non-allergic pediatric patient." to result.rationale<br>Rationale: pediatric first-line                                       |   |   | ✓ |

## 3020 · Determine\_Dose

### FIRST

| #      | Condition                                                                                                                              | 1 | 2 | 3 |
|--------|----------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 1      | patient.pcr > constants.renal_pcr_threshold and result.ccr < constants.renal_ccr_threshold<br>Renal impairment: PCr > 1.4 and CCr < 50 | Y | – | – |
| 2      | patient.age >= constants.dose_low_age and patient.age <= constants.dose_high_age<br>Standard-dose age band (15..60)                    | – | Y | N |
| ACTION |                                                                                                                                        | 1 | 2 | 3 |
| 1      | set result.dose_mg = constants.reduced_dose<br>Renal-reduced dose                                                                      | ✓ |   |   |
| 2      | set result.dose_mg = constants.standard_dose<br>Standard dose for 15..60                                                               |   | ✓ |   |
| 3      | set result.dose_mg = constants.reduced_dose<br>Reduced dose for <15 or >60                                                             |   |   | ✓ |
| 4      | set result.renal_adjustment = true<br>Flag the renal adjustment                                                                        | ✓ |   |   |
| 5      | set result.frequency_hours = constants.dosing_interval_hours<br>Dosing interval (hours)                                                | ✓ | ✓ | ✓ |
| 6      | set result.duration_days = constants.therapy_duration_days<br>Therapy duration (days)                                                  | ✓ | ✓ | ✓ |

## 4000 · Check\_Drug\_Interactions

FIRST

contexts:  
for all patient.active\_medications

| #      | Condition                                                                                                                                                                                       | 1 | 2 | 3 | 4 | 5 | 6 |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|
| 1      | result.recommended_drug is equal to ignore case "Levofloxacin"<br>Recommended drug is Levofloxacin                                                                                              | Y | Y | Y | - | - | - |
| 2      | result.recommended_drug is equal to ignore case "Amoxicillin"<br>Recommended drug is Amoxicillin                                                                                                | - | - | - | Y | Y | - |
| 3      | result.recommended_drug is equal to ignore case "Cefuroxime"<br>Recommended drug is Cefuroxime                                                                                                  | - | - | - | - | - | Y |
| 4      | medication.name is equal to ignore case "Coumadin" or medication.name is equal to ignore case "Warfarin"<br>Active med is an anticoagulant                                                      | Y | - | - | Y | - | - |
| 5      | medication.name is equal to ignore case "Theophylline"<br>Active med is theophylline                                                                                                            | - | Y | - | - | - | - |
| 6      | medication.name is equal to ignore case "NSAID" or medication.name is equal to ignore case "Ibuprofen" or medication.name is equal to ignore case "Naproxen"<br>Active med is an NSAID          | - | - | Y | - | - | - |
| 7      | medication.name is equal to ignore case "Methotrexate"<br>Active med is methotrexate                                                                                                            | - | - | - | - | Y | - |
| 8      | medication.name is equal to ignore case "Probenecid"<br>Active med is probenecid                                                                                                                | - | - | - | - | - | Y |
| ACTION |                                                                                                                                                                                                 | 1 | 2 | 3 | 4 | 5 | 6 |
| 1      | add "Levofloxacin may potentiate the anticoagulant effect of warfarin/Coumadin, raising INR and bleeding risk. Monitor coagulation closely." to result.warnings<br>Levofloxacin + anticoagulant | ✓ |   |   |   |   |   |
| 2      | add "Levofloxacin may increase theophylline plasma levels; monitor for theophylline toxicity." to result.warnings<br>Levofloxacin + theophylline                                                |   | ✓ |   |   |   |   |
| 3      | add "Concomitant NSAIDs with Levofloxacin may increase the risk of CNS stimulation and seizures." to result.warnings<br>Levofloxacin + NSAID                                                    |   |   | ✓ |   |   |   |
| 4      | add "Amoxicillin may enhance the anticoagulant effect of warfarin/Coumadin; monitor INR." to result.warnings<br>Amoxicillin + anticoagulant                                                     |   |   |   | ✓ |   |   |
| 5      | add "Amoxicillin may reduce renal clearance of methotrexate, increasing methotrexate toxicity." to result.warnings<br>Amoxicillin + methotrexate                                                |   |   |   |   | ✓ |   |

| # | Condition                                                                                                                               | 1 | 2 | 3 | 4 | 5 | 6 |
|---|-----------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|
| 6 | add "Probenecid reduces renal excretion of Cefuroxime, raising its plasma concentration." to result.warnings<br>Cefuroxime + probenecid |   |   |   |   |   | ✓ |